

Phase 2 - Identity Signal

Event and Scoring

This challenge is released in **three cumulative phases**.

Current Phase	Evaluation Includes
Phase 1	Phase 1 requirements
Phase 2	Phases 1 and 2 requirements
Phase 3	Phases 1, 2 and 3 requirements

Each phase's evaluation re-tests the requirements of every earlier phase within the same evaluation run. A Phase 2 evaluation therefore re-checks Phase 1 behaviour, and a Phase 3 evaluation re-checks Phases 1 and 2 behaviour. Extend your system without breaking what already works.

Phases Schedule

Start with Phase 1 only. Phases 2 and 3 are unlocked during the event — ignore them until announced. When they arrive, extend your system without breaking what already works.

Scoring Overview

You do **not** need to match an absolute "correct" score for each transaction. You need a coherent model that consistently ranks suspicious activity above ordinary flow.

Submissions are evaluated across two dimensions:

- **Detection Quality:** How accurately your system ranks more suspicious transactions above less suspicious ones.
- **Structural Consistency:** How coherently your model behaves across structurally related scenarios. Systems built on principled graph models are expected to outperform implementations tuned to specific patterns.

In addition, evaluating a phase during its **active window** grants an **earliness bonus**. The bonus rewards a working system evaluated early within a phase's window; it is applied on top of the two scored dimensions and does not change what is scored.

Final standing reflects the two scored dimensions combined, with the per-phase earliness bonus applied.

Phase start times are announced via the coordinator and the challenge room / Teams channel; they are not published in advance in these documents.

System Requirements

Service Overview

Participants must implement a real-time **transaction risk scoring service** for an anti-money laundering (AML) environment.

The system receives transactions continuously and must assign a **risk score between 0.0 and 1.0** to each transaction at the moment it is processed.

The service is expected to operate under **stateful, streaming conditions**, where historical transactions within the active lookback window influence future scoring behavior.

Required Endpoints

Participants must implement the following HTTPS endpoints:

Health Check

```
GET /ghost-chains/health
```

```
{ "status": "ok" }
```

This endpoint is used to verify service availability.

State Reset

POST /ghost-chains/reset

Request:

```
{ "clearTransactions": true }
```

Response:

```
{ "clearTransactions": true }
```

Behavior:

- Clears all internal state related to previously processed transactions.
- Resets graph construction, caches, and any derived structures.
- Must restore the system to a clean initial state equivalent to startup.

Transaction Processing

POST /ghost-chains/transactions

Request fields:

- transactions : Array of transaction objects:
 - txId : Unique string identifier for the transaction.
 - fromUserId : Identifier of the sending entity. (User is a convenience label for any identity — account, legal entity, or other counterparty.)
 - toUserId : Identifier of the receiving entity.
 - amount : Transfer amount as a number.
 - createdAt : ISO 8601 timestamp of the transaction.

- `ipAddress` (*optional*): Network address used to initiate the transaction. Omitted when unknown.
- `deviceId` (*optional*): Device identifier used to initiate the transaction. Omitted when unknown.

Optional fields may be absent on any transaction; this must not cause processing to fail.

Request:

```
{
  "transactions": [
    {
      "txId": "tx_meridian_001",
      "fromUserId": "meridian_holdings",
      "toUserId": "apex_logistics",
      "amount": 370.0,
      "createdAt": "2026-06-08T12:00:00Z"
    },
    {
      "txId": "tx_cascade_014",
      "fromUserId": "cascade_payments",
      "toUserId": "horizon_capital",
      "amount": 100.0,
      "createdAt": "2026-06-08T12:01:00Z"
    }
  ]
}
```

Response fields:

- `transactions` : Array of result objects:
 - `txId` : Echoes the transaction identifier from the request.
 - `riskScore` : Risk score in [0.0, 1.0].

Response:

```
{
  "transactions": [
    { "txId": "tx_meridian_001", "riskScore": 0.0 },
    { "txId": "tx_cascade_014", "riskScore": 0.0 }
  ]
}
```

```
}
```

Behavior:

- Each transaction must be assigned a risk score immediately upon processing.
- Multiple transactions in a single request must be processed sequentially in order.
- The response must preserve input ordering.

Getting Started

1. Implement the three endpoints above.
2. Smoke-test with the commands below. Replace `localhost:8080` with your server's host address.
3. Register your public base URL with the coordinator and trigger an evaluation.

```
curl -s http://localhost:8080/ghost-chains/health
```

```
curl -s -X POST http://localhost:8080/ghost-chains/reset \
-H 'Content-Type: application/json' \
-d '{"clearTransactions": true}'
```

```
curl -s -X POST http://localhost:8080/ghost-chains/transactions \
-H 'Content-Type: application/json' \
-d '{
  "transactions": [
    {
      "txId": "tx_meridian_001",
      "fromUserId": "meridian_holdings",
      "toUserId": "apex_logistics",
      "amount": 370.0,
      "createdAt": "2026-06-08T12:00:00Z"
    }
  ]
}'
```

State and Execution Model

Build a stateful, streaming service that maintains a rolling graph of account transactions and assigns relative risk scores within a bounded lookback window.

Streaming state. Score each transaction using only information available at that point. State is updated incrementally — no reprocessing of history is required. Maintain state sufficient for:

- Graph relationships between entities
- Temporal ordering of transactions
- Derived structural and behavioural signals

Lookback window (W = 24 hours). Only transactions created within the most recent 24 hours are active. Expired transactions must be removed from graph state and must not influence scoring. Be precise about boundary conditions.

Ordering. Within a single request, process transactions in the order provided. Across requests, arrival order defines global state evolution.

Idempotency. Each `txId` is unique. If a `txId` is submitted more than once with an identical payload, return the original score and make no state changes. Consider what should happen if the payload differs.

Risk scores are real numbers in **[0.0, 1.0]** representing relative suspiciousness — not calibrated probabilities. A higher score means more suspicious. Scores must be comparable within the same running system state, and for identical inputs after a reset, outputs must be consistent.

Forward compatibility. Later phases introduce new optional fields. Ignore unknown or absent fields gracefully; do not reject transactions that include unrecognised attributes.

Performance. Handle continuous streaming input with memory usage bounded by the active lookback window.

System Observation and Diagnostics

When your submission produces a ranking that disagrees with the reference model, the platform does not disclose absolute scores or internal evaluation details. Instead, a diagnostic array is returned identifying which signal dimensions showed disagreement. A phase's evaluation can emit only the observation categories listed in that phase's document.

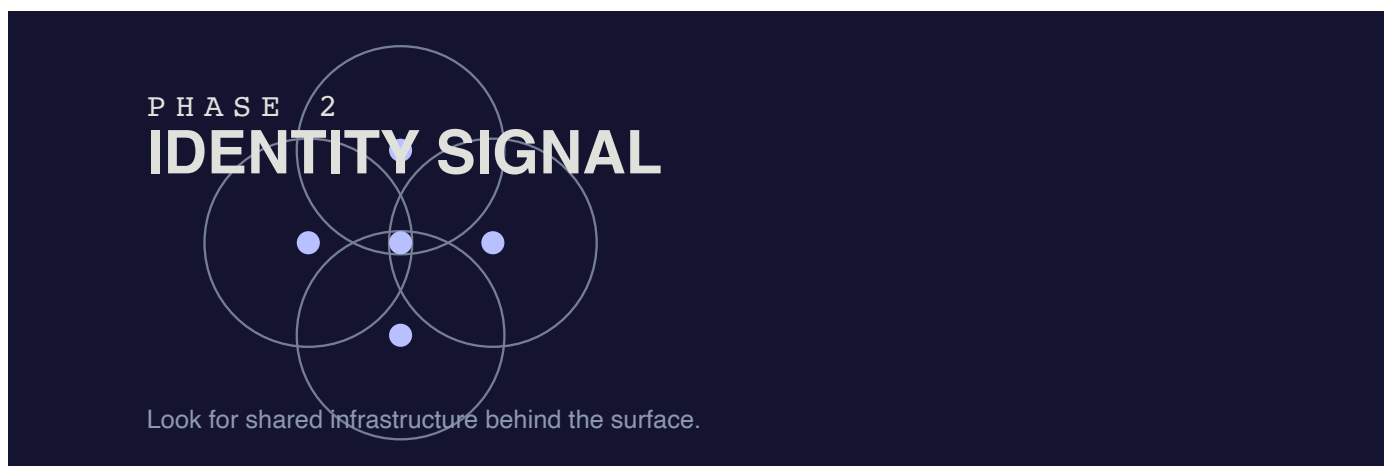
Severity Interpretation

Observation categories describe where evaluator disagreement was detected. Multiple categories may be reported simultaneously. Severity reflects the magnitude of disagreement between your submission's output and reference behaviour for the evaluated scenario. Severity is computed dynamically and is not tied to any specific challenge phase or test difficulty level. If no disagreement is detected for a scenario, no diagnostic payload is generated.

Diagnostic Payload Format

`STRUCTURAL_DEVIATION: Moderate, TEMPORAL_DEVIATION: Low`

Phase 2 Briefing



Coordinated financial networks often share underlying infrastructure. Transactions that look unrelated on the graph may share a network address or device — a hint of common control.

A single shared attribute can be coincidence (office Wi-Fi, cloud NAT). When identity lines up with structural flow — or the same identity appears across disconnected components — treat it as a stronger combined signal.

Assign a **higher risk score** when identity evidence increases combined suspicion.

Core Principle

Optional fields `ipAddress` and `deviceId` contribute an identity signal relative to where

the transaction sits in the active graph. When both are present, treat them as independent dimensions.

Shared identity across **disconnected** components is a distinct coordination hint — not automatic proof of risk on its own.

Missing identity on a connected path. When an identity attribute that appeared on earlier legs of a continuous flow is absent on a later leg, the absence itself can be a signal: dropping a network address or device identifier mid-path is a way to break the trail. Missing fields are normal on unrelated transactions; the suspicious case is a *consistent flow that stops carrying its identity*. Weigh absence against the surrounding structure rather than treating every missing field as suspicious.

Objectives

- Combine identity scoring with structural scoring
- Tolerate missing identity fields

Earlier Phases

All Phase 1 requirements continue to apply in Phase 2. The Phase 1 Constraints Checklist still applies. Phase 2 introduces no new mechanical requirements; it activates `ipAddress` and `deviceId` as identity signals (see Core Principle). Behaviours not covered here are left for you to reason about.

Phase 2 Examples

These examples show transaction sequences from first to last. Assume that the preceding transactions have already been scored, and that the final transaction is now being evaluated. These examples show how evidence changes — they do **not** define a strict risk ordering between scenarios.

Example 1 - Consistent Identity

1. Meridian Holdings → Apex Logistics (`deviceId` : `dev_ios_7f3a91`)
2. Apex Logistics → Cascade Payments (`deviceId` : `dev_ios_7f3a91`)
3. Cascade Payments → Horizon Capital (`deviceId` : `dev_ios_7f3a91`)

Interpretation:

- A single directed flow carries a consistent device identifier throughout. No identity anomaly exists within this segment.

Example 2 - Identity Divergence Under Branching

1. Meridian Holdings → Apex Logistics (deviceId : dev_ios_7f3a91)
2. Apex Logistics → Cascade Payments (deviceId : dev_ios_7f3a91)
3. Apex Logistics → Sterling Bridge (deviceId : dev_ios_7f3a91)
4. Cascade Payments → Oakridge Imports (deviceId : dev_android_c2e4b8)

Interpretation:

- Two branches extend from Apex Logistics. One branch introduces a new device identifier on Cascade Payments → Oakridge Imports. Device dev_ios_7f3a91 is no longer uniform across the full reachable subgraph from Meridian Holdings.

Example 3 - Identity Shift Mid-Flow

1. Meridian Holdings → Apex Logistics (deviceId : dev_ios_7f3a91)
2. Apex Logistics → Cascade Payments (deviceId : dev_ios_7f3a91)
3. Cascade Payments → Horizon Capital (deviceId : dev_android_c2e4b8)
4. Horizon Capital → Nimbus Trading (deviceId : dev_android_c2e4b8)

Interpretation:

- A structurally continuous path Meridian Holdings → Apex Logistics → Cascade Payments → Horizon Capital → Nimbus Trading exists. The device identifier changes at the Cascade Payments → Horizon Capital transition, weakening the confidence that a single identity cluster explains the full path. The structural relationship between entities remains valid; identity and structural observations must be considered together rather than in isolation.

Example 4 - Shared Identity Across Disconnected Components

1. Meridian Holdings → Apex Logistics (ipAddress : 10.0.0.1)
2. Cascade Payments → Horizon Capital (ipAddress : 10.0.0.1)
3. Oakridge Imports → Sterling Bridge (ipAddress : 10.0.0.1)

Interpretation:

- Three transactions share a network address with no structural connectivity between their participants. This creates a potential identity relationship between entities that is not visible from graph structure alone. Shared network infrastructure may indicate coordination, but may also arise from legitimate network aggregation. Structural or value-flow evidence from the same components may be required to determine the significance of this identity signal.

Signal Relationships

These examples illustrate how identity observations change the available evidence about the active transaction graph. They are not intended to establish a direct risk ordering.

- **Example 1** (identity agreement): a single structural flow carries a consistent identity signal throughout. Structural and identity observations reinforce each other within this segment.
- **Example 2** (identity divergence at a branch): the identity signal changes at one outgoing edge. The two branches now carry different identity evidence, and neither independently characterises the full reachable subgraph.
- **Example 3** (identity disagreement within a continuous flow): the structural path remains unbroken, but the identity evidence changes partway through. Both observations are valid and must be weighed together rather than in isolation.
- **Example 4** (identity reuse across disconnected components): the same identity signal appears in multiple unrelated components. This creates a potential cross-structural relationship not visible from graph structure alone, but does not independently establish risk.

Phase 2 Diagnostics Vocabulary

Phase 2 evaluations can emit the following observation categories:

- `STRUCTURAL_DEVIATION` : Disagreement detected in the evaluation of structural signals.
- `TEMPORAL_DEVIATION` : Disagreement detected in temporal signal evaluation or lookback window handling.
- `IDENTITY_DEVIATION` : Disagreement detected in the evaluation of identity signals.